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Studierichting: Informatica
Oefeningenreeks 5G nr. 6.2

$f(x) = \sqrt{x}$

$g(x) = 2\sqrt[4]{x}$

$f(x) = g(x)$

$\sqrt{x} = 2\sqrt[4]{x}$

$\Leftrightarrow x^{\frac{1}{2}} = 2x^{\frac{1}{4}}$

$\Leftrightarrow x^{\frac{1}{2}} - 2x^{\frac{1}{4}} = 0$

$\Leftrightarrow x^{\frac{1}{4}} \left(x^{\frac{1}{4}} - 2\right) = 0$

$\Leftrightarrow x^{\frac{1}{4}} = 0$ of $x^{\frac{1}{4}} = 2$

$\Leftrightarrow x = 0$ of $x = 16$

$\Rightarrow x \in [0, 16]$

We testen een punt tussen 0 en 16: $x = 1$

$f(1) = \sqrt{1} = 1$

$g(1) = 2\sqrt[4]{1} = 2$

$\Rightarrow g(x)$ is de bovenste functie

$\Rightarrow V = \pi \int_0^{16} \left(g(x)^2 - f(x)^2\right) dx$

$= \pi \int_0^{16} \left((2\sqrt[4]{x})^2 - (\sqrt{x})^2\right) dx$

$(2\sqrt[4]{x})^2 = 4x^{\frac{1}{2}}$

$(\sqrt{x})^2 = x$

$\Rightarrow (2\sqrt[4]{x})^2 - (\sqrt{x})^2 = 4x^{\frac{1}{2}} - x$

$\Rightarrow V = \pi \int_0^{16} \left(4x^{\frac{1}{2}} - x\right) dx$

$= \pi \left[\frac{4x^{\frac{3}{2}}}{\frac{3}{2}} - \frac{x^2}{2}\right]_0^{16}$

$$= \pi \left[\frac{8}{3} x^{\frac{3}{2}} - \frac{x^2}{2} \right]_0^{16}$$

$$= \pi \left(\frac{8}{3} (16)^{\frac{3}{2}} - \frac{16^2}{2} \right) - \pi \left(\frac{8}{3} (0)^{\frac{3}{2}} - \frac{0^2}{2} \right)$$

$$16^{\frac{3}{2}} = \left(\sqrt{16} \right)^3 = 4^3 = 64$$

$$\Rightarrow V = \pi \left(\frac{8}{3} \cdot 64 - \frac{256}{2} \right) - \pi(0)$$

$$= \pi \left(\frac{512}{3} - 128 \right)$$

$$= \pi \left(\frac{512}{3} - \frac{384}{3} \right)$$

$$= \pi \cdot \frac{128}{3}$$

$$\Rightarrow V = \frac{128\pi}{3}$$

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References